

Online Complaint Registration and Management System

Phase 2: Requirement Analysis

SmartBridge Internship Project Submission

Team: Online Complaint Registration and Management System

2.1 Customer Journey Map

Stage	User Action	Touchpoint	Pain Point Addressed
Awareness	Discovers the complaint portal	Landing page (Home)	No prior visibility into how to raise an issue
Registration	Creates an account	Register page	No account/record tied to past complaints
Complaint Submission	Fills complaint form (title, category, location, priority)	Add Complaint page	Complaints previously scattered across calls/paper
Tracking	Views status on dashboard	Dashboard / Complaint Detail page	No visibility into progress previously
Resolution	Receives status change to Resolved	Notifications, Complaint Detail page	Long, silent wait times with no closure

2.2 Solution Requirements

2.2.1 Functional Requirements

ID	Requirement	Description
FR-1	User Registration & Login	Users can create an account and log in securely using email and password (JWT-based authentication).
FR-2	Role-Based Access	System supports three roles — User, Agent, Admin — each with distinct permissions.
FR-3	Complaint Registration	Users can submit a complaint with title, description, category, location, and priority.
FR-4	Complaint Tracking	Users can view the real-time status of their submitted complaints.
FR-5	Complaint Assignment	Admins can assign complaints to available agents.
FR-6	Status Management	Agents/Admins can update complaint status (Pending → Assigned → In Progress → Resolved → Closed).
FR-7	Notifications	Users receive notifications when their complaint status changes or an agent is assigned.
FR-8	Search & Filter	Users/Admins can search and filter complaints by status, category, or keyword.
FR-9	Admin Analytics Dashboard	Admins can view aggregate statistics: total, pending, resolved complaints, and category/status breakdowns.
FR-10	User Management	Admins can view all users, change roles, and remove accounts.

2.2.2 Non-Functional Requirements

ID	Requirement	Description
NFR-1	Security	Passwords hashed with bcrypt; JWT tokens for stateless authentication; role-based authorization on all protected routes.
NFR-2	Usability	Responsive UI (Tailwind CSS) accessible on desktop and mobile browsers.
NFR-3	Performance	API responses under 500ms for standard CRUD operations under normal load.
NFR-4	Scalability	MongoDB Atlas cloud database supports horizontal scaling as complaint volume grows.
NFR-5	Availability	Hosted on Render with backend/frontend deployed as independent services for independent scaling and uptime.
NFR-6	Maintainability	Modular MVC-style backend (routes/controllers/models) and component-based React frontend.

2.3 Data Flow Diagrams

2.3.1 DFD Level 0 (Context Diagram)

The context diagram shows the system as a single process interacting with three external entities — User, Agent, and Admin — and the underlying database.

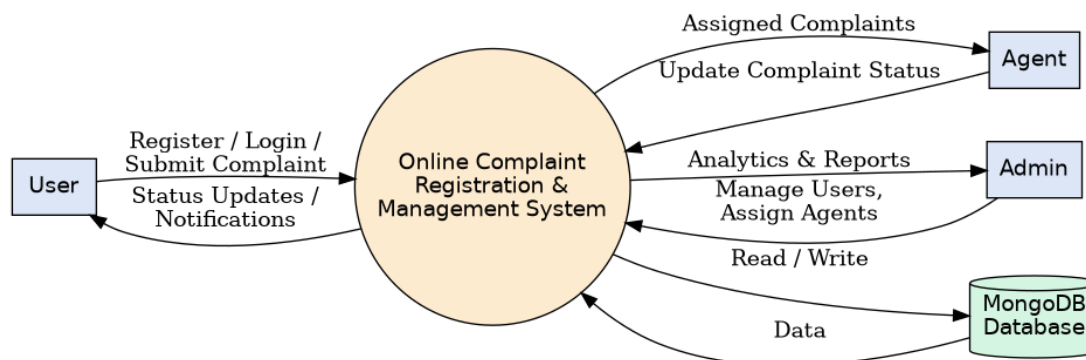


Figure 2.1 — DFD Level 0: Context Diagram

2.3.2 DFD Level 1

The Level 1 diagram decomposes the system into its core processes: Register/Login, Submit Complaint, Track Complaint, Assign & Update Status, and Notify User — along with the data stores they interact with.

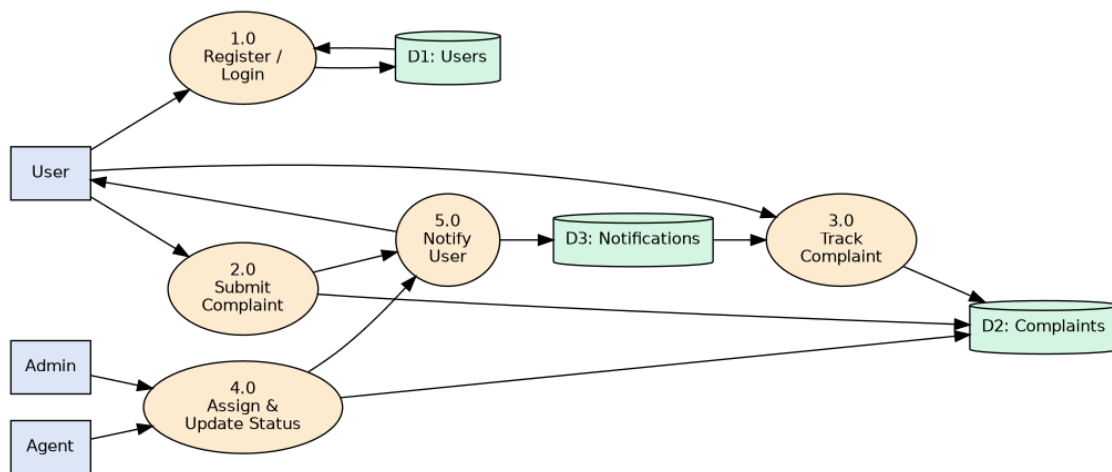


Figure 2.2 — DFD Level 1: Process Decomposition

2.4 Technology Stack

Layer	Technology
Frontend	React.js (Vite), Tailwind CSS, React Router DOM, Axios
Backend	Node.js, Express.js
Database	MongoDB Atlas (Mongoose ODM)
Authentication	JSON Web Tokens (JWT), bcryptjs for password hashing
Deployment	Render (Web Service for backend, Static Site for frontend)
Version Control	Git & GitHub